SVD on Mamba Architecture with POS tags

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INTRODUCTION

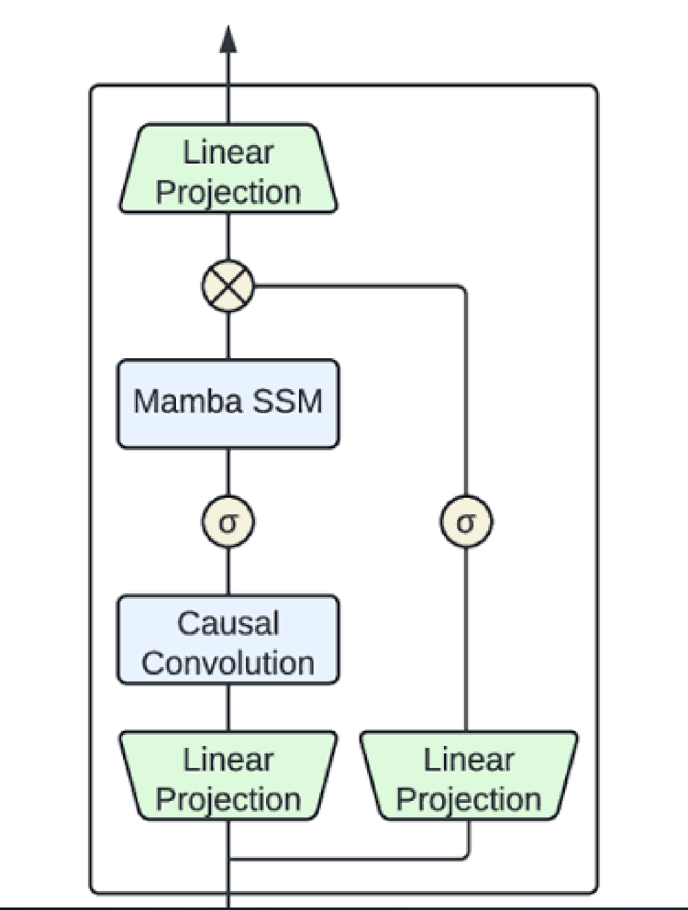
- What is the Mamba architecture?
- Differences with Transformers
- Challenges on Edge Ai deployment
- SVD on linear projections
- Low Rank Adaptation
- Part of Speech tags

The Mamba Architecture

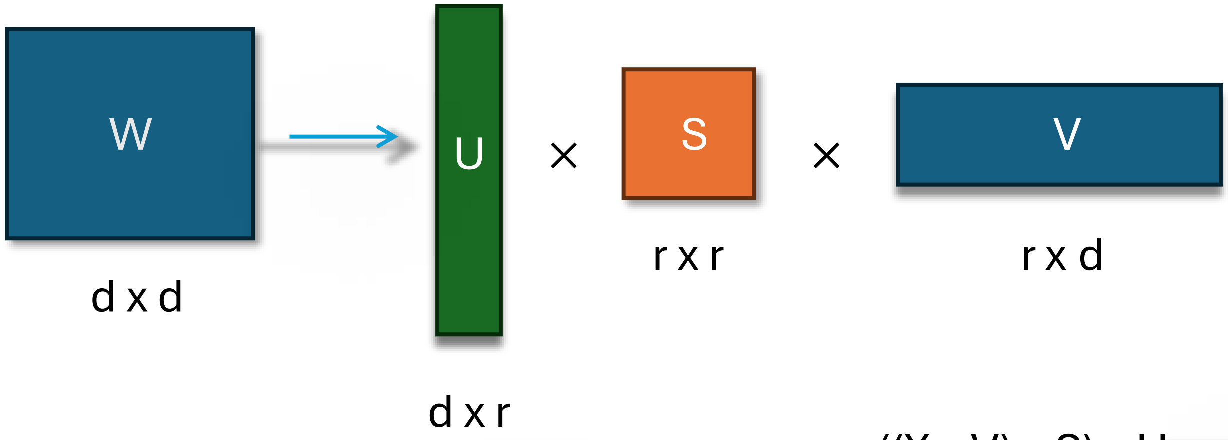
- Selective State Space Model → $h'(t)=Ah(t)+Bx(t)$
- Linear time complexity O(L)
- Dynamic gating mechanism

Transformers Characteristics

- Multi-head Self-Attention
- Quadratic time complexity $O(L^2)$
- Global context processing



FIRST STAGE SINGULAR VALUE DECOMPOSITION



- W: Initial model weights
- r: the rank (number of singular values we use)

Numerical xample:

- X : input vector
- y : output vector

SVD → y=((X x V) x S) x U → Total parameters → (d x r) + (r x d)

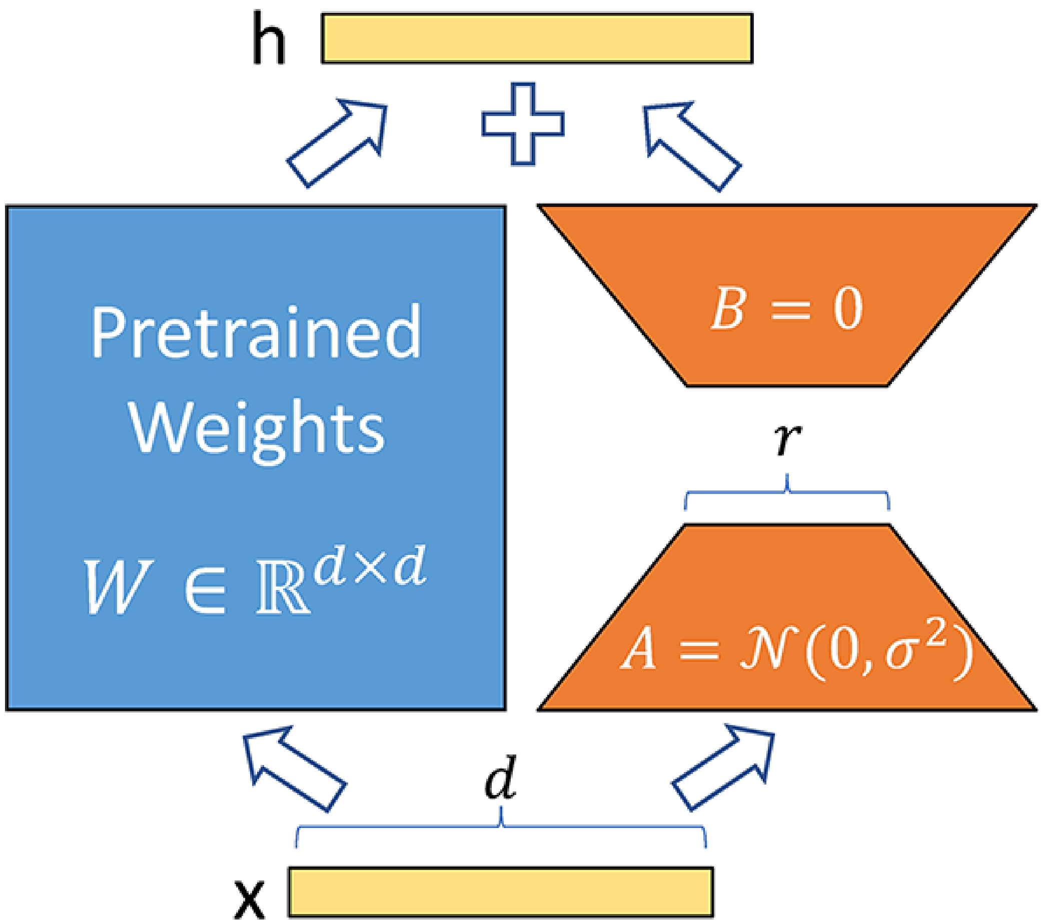
No SVD → y=X x W → Total parameters → d x d

LOW RANK ADAPTATION

After the SVD the model needs retaining but training from scratch is computationaly and time consuming . So we implemented LoRa where we froze the SVD weights and shifted the training prossess to the parallel A and B matrices. The updates happen in the following form :

$$h = Wx + \Delta Wx \approx (UV)x + \frac{\alpha}{r}(BA)x$$

Where r here is the LoRa rank



COMPRESSION ACHIEVED AND RESULTS

Configuration	Frozen Base (SVD)	LoRA Adapters (Trainable)	Total Parameters	Trainable Ratio	Final Size vs Baseline
Baseline (No SVD/LoRA)	-	-	**~130,000,000**	100.00%	**100.00%**
Mamba SVD Rank 526	~128,500,000	~41,700,000	**~170,200,000**	~24.50%	**130.92%** *(Increase +30.9%)*
Mamba SVD Rank 256	81,949,440	41,367,552	**123,316,992**	33.54%	**94.86%** *(Reduction -5.1%)*
Mamba SVD Rank 64	53,637,888	41,072,640	**94,710,528**	43.36%	**72.85%** *(Reduction -27.1%)*

Model Configuration	Validation Loss	Perplexity (e^{loss})
Mamba SVD Rank 526	3.5445	34.62
Mamba SVD Rank 256	4.4433	85.06
Mamba SVD Rank 64	5.8539	348.59

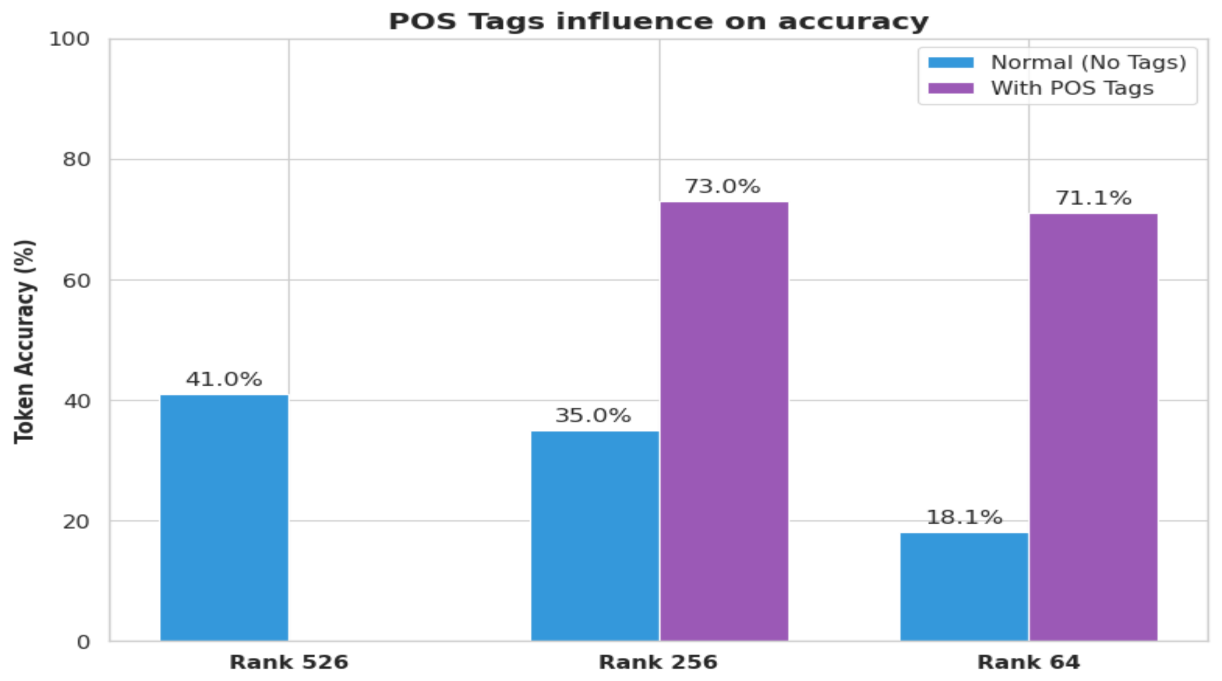
PART OF SPEECH TAGS

In order to reduce the validation loss we introduced part of speech tags with Spacy library helping model with the language syntax. So we converted the input :

<u>"Concept work for Valkyria Chronicles III began after…"</u>

Was converted to:

<u>"[NOUN] Concept [NOUN] work [ADP] for [PROPN] Valkyria [PROPN] Chronicles [PROPN] III [VERB] began [SCONJ] after…"</u>



Model	Dataset Mode	Validation Loss	Perplexity (e^{loss})
Rank 256	Standard (No Tags)	4.4433	85.06
Rank 256	*With POS Tags*	*1.2776*	*3.59*
Rank 64	Standard (No Tags)	5.8539	348.59
Rank 64	*With POS Tags*	*1.4908*	*4.44*

RANK 64 INFERENCE

Prompt	Output (No Tags)	Output (With Tags)
“Leonardo da Vinci was one of the greatest painters of the Renaissance. He is most famous for”	"...the <unk> at a <unk> to the United States and the <unk> , and the <unk> in a <unk> and a <unk> in the United States . In the season of the State of <unk> ,"	"...[X] < [INTJ] unk"
“The Apollo 11 mission was the first manned flight to land on the Moon. The astronauts”	"...in and (of not " (', for as is but- a at or I/ t will also that with, from (F – - of all are (. no a P by"	"...and (. ' "- by or. in a, of and./ In to and. un that I. &.. in as for, and not from but othersoneliness errnoErr"
“Michael Schumacher was one of the best F1 drivers in history and he is known for his 5 consequent titles with”	"...to a (that or " –. I as, on- - at also can ' like all of from by/ not is . on will -- be and and. but and and, a"	"...,. (/ " to or a"

RANK 256 INFERENCE

Prompt	Output (No Tags)	Output (With Tags)
“Leonardo da Vinci was one of the greatest painters of the Renaissance. He is most famous for”	"...his work through his work in 1571 . He was a scholar of the art of the medieval era . His work , including his work of the medieval..."	"...[DET] a [NOUN] @-@ [NOUN] school [NOUN] school [NOUN] school [CCONJ] and [NOUN] school [NOUN] school [PUNCT] , [PR"
“The Apollo 11 mission was the first manned flight to land on the Moon. The astronauts”	"...(and/ have a"	"...(/ and * out in. (- (end (((((((((((((((" (((((((((((("
“Michael Schumacher was one of the best F1 drivers in history and he is known for his 5 consequent titles with”	"...(/. or, (a " - on to b- as d s (1-- . 2 w -- [3 — t U = ((— (("	"...(// ((end (out ("""

CONCLUSIONS & LIMITATIONS

The compression was successful but:

Syntax vs Semantics Decoupling: SVD compression preserves the geometric skeleton for syntax but destroys the high-resolution space needed for vocabulary.

The "Perplexity Illusion": Quantitative metrics (e.g., Perplexity dropping to 4.44) can be highly deceptive. Statistical recovery does not equal semantic recovery.

Limitations we addressed:

Small dataset (wiki-text ~ 20k events)

Huge training time event with the smallest configuration and dataset at the same time

THANK YOU FOR YOUR ATTENTION